

POTPRESS · REQUIREMENTS

PotPress requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; R10 reported against the target
0.8	2026-09-30	Amish Chadha	Status from PPR-CAL-001 v0.6 after the design for construction (PPR-DDR-004); R10 not met on paper; no requirement text changed
0.7	2026-09-27	Amish Chadha	Cost overrun decided by Amish on 2026-09-27; R10 target \$1,060, status met on paper
0.6	2026-09-26	Amish Chadha	Guarded version (decided by Amish 2026-09-26)
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish; R10 target \$990, status met on paper
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R3 at 10 t with a metal stop, R9 pin interlock, R10 \$930, R11 rack 0.8 x 0.8 m and folded rails, R12 lead-free scrap; status from PPR-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Apply Amish's 2026-09-25 decisions (PPR-DDR-001); R10 target \$720, R6 band decided; status from PPR-

CAL-001; proposed changes to R3 and R11

0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

PotPress requirements

These requirements were checked by calculation at TRL 3 (PPR-CAL-001 v0.6). They are not yet validated with a filter factory and will be revised after co-design sessions (see PPR-PRB-001). With the design made constructable (PPR-DDR-004, 2026-09-30), one is over the value-engineering target (R10, the parts cost of \$1,143 against the \$1,060 target, \$83 over), two are at risk (R2 and R7) and nine are met on paper; see Table 2. Amish's decisions of 2026-09-25 are recorded in PPR-DDR-001 and PPR-DDR-002: the R6 default band, R3 judged at the 10 t working force, the R10 target of \$930 (topped up to \$990 by Amish on 2026-09-26 and raised to \$1,060 by Amish on 2026-09-27, PPR-DDR-003), the R11 rack area of 0.8 x 0.8 m, the pin-presence interlock in R9 and lead-free scrap in R12. The guarded version and the guarding detail in R9 were decided by Amish on 2026-09-26 (PPR-DDR-003).

The **reference filter** used throughout is the common flowerpot form: inner rim diameter 280 mm, inner depth 240 mm, wall 15 mm, flat rim about 345 mm across, about 12 L to the brim and about 10 L working volume (dimensions are estimates based on the 280 by 250 mm form reported by [Potters for Peace](#)).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Form the reference filter	Inner rim 280 mm, inner depth 240 mm, wall 15 mm, flat rim; geometry set by parameters in one source file	Model check; later measurement of pressed pots
R2	Even wall	Wall thickness within ± 1 mm around the circumference and from pot to pot; molds self-center to within 0.5 mm	Tolerance stack and guide design; later sectioned pots
R3	Pressing force and frame strength	20 t (196 kN) rated jack; frame, pin, bolts and platen carry 1.5 times the jack rating (294 kN) without yield; deflection between the molds under 1 mm at the 10 t maximum working force, with the molds closing on a metal stop so frame stretch does not set the wall (decided by Amish, 2026-09-25)	Beam and pin calculation; later proof load
R4	Throughput	Press cycle of 6 min or less with two operators, so 50 or more pots per 6 h of pressing	Cycle breakdown; later timed trial
R5	Mold opening	Male mold lifts at least 30 mm clear of the female rim so the mold carriage slides out for loading and demolding; pressing stroke within one jack stroke	Kinematic layout
R6	Flow-rate QC	Rack tests 4 soaked filters at once; T-gauge reads flow over 1 h to 0.1 L/h; repeatability ± 0.1 L/h; acceptance band set per factory (default 1.0 to 2.5 L/h in the first hour, corrected to 25 °C; decided by Amish, 2026-09-25)	Gauge calibration by volume; later repeat tests
R7	Molds made locally	Molds cast from 3D-printed patterns printed on a 250 x 250 mm class printer; finishing with hand tools and a drill press; no lathe over 300 mm swing	Pattern split and finishing review with a foundry
R8	Frame built locally	Standard steel channel and plate; stick welding, drilling and bolting only; heaviest single part 40 kg or less for two-person handling	Part list and mass estimate
R9	Safe operation	Fixed welded-mesh guards on the sides, back and roof and a hinged front gate enclose every pinch point of the molds, platen, jack and carriage; guard openings and their distance from moving parts prevent finger reach into the pinch zone, judged against ISO 13857; a guard-locking gate interlock stops the jack release closing (so the jack cannot be pumped to build pressure) unless the gate is shut, and keeps the gate shut while the release is closed; jack pump and release worked from outside the guard; a pin-presence interlock prevents pressing unless the load pin is fully home (pin interlock decided by Amish, 2026-09-25; guarding decided by Amish, 2026-09-26)	Hazard review and guard layout in the model; later guard check against ISO 13857
R10	Affordable	Press plus QC rack at or under the \$1,060 value-engineering target in parts (a hypothetical control target; raised from \$600 to \$720 and then to \$930 by Amish, 2026-09-25; topped up to \$990 by Amish, 2026-09-26; raised to \$1,060 by Amish, 2026-09-27)	Priced BOM (bom/ bom.csv)
R11	Footprint	Press within 1.0 x 0.7 m floor area and 2.0 m height, with the rail extension folded; QC rack within 0.8 x 0.8 m (relaxed from 0.8 x 0.5 m by Amish, 2026-09-25)	Model check
R12	Product-safe materials	Faces that touch clay or test water are aluminum, food-grade polyethylene or stainless steel; no lead-based paint or oiled release agents on mold faces; molds cast from lead-free scrap (no free-machining alloys; decided by Amish, 2026-09-25)	Material list review

Table 2. Status against each requirement at TRL 3 (from PPR-CAL-001 v0.6 and, for R9, R10 and R11, the guarded model and BOM; least certain first).

ID	TRL 3 value	Status
R10	Priced BOM \$1,143 (press \$1,060, QC rack \$83) with the constructable design (PPR-DDR-004) against \$1,060	Over the value-engineering target by \$83 (7.8 %)
R2	Wall ± 1.40 mm as cast; ± 0.44 mm with molds hand-finished to templates; coaxial ± 0.44 mm via match-drilled dowels	At risk
R7	16 flat-back pattern segments for a 250 mm printer, no core boxes; no lathe needed; cavity hand finishing unproven	At risk
R9	Guards, gate and interlocks modeled (PPR-DDR-003): 12.7 mm welded mesh with nearest moving parts about 105 to 110 mm behind it; guard-locking gate interlock and pin-presence plunger on the jack release lock bar; pump handle through a 30 mm slot, release knob outside	Met on paper; openings and distances assumed against ISO 13857, not checked (proposed, awaiting Amish)
R1	Reference filter from PARAMS in <code>cad/src/model.py</code> : 12.30 L to the brim, 9.91 L working	Met on paper
R3	No yield at 294 kN: beams 190 MPa, uprights 54 MPa, 60 mm pin 272 MPa (yield 650), M16 joint bolts 117 MPa, male flange on the stop 62 MPa. Deflection between the molds 0.66 mm at 10 t (1.97 mm at 294 kN)	Met on paper
R4	5.2 min cycle, 69 pots per 6 h	Met on paper, thin margin
R5	310 mm opening against 270 mm needed; 110 of 150 mm jack stroke	Met on paper
R6	4 stations; 0.1 L is 1.69 mm on the gauge; temperature correction to 25 °C	Met on paper; repeatability not verifiable at TRL 3
R8	Heaviest part 39.6 kg (platen); upright joints bolted, largest frame part 38.0 kg	Met on paper
R11	Press 940 x 700 mm guarded with the rail extension folded (the pump handle stands 158 mm outside the right guard while in use), 1,806 mm tall; QC rack 780 x 780 mm	Met on paper, at the 0.7 m depth limit with no margin
R12	Aluminum faces, polyethylene liners, HDPE buckets; lead-free scrap alloy	Met on paper

Requirement changes decided by Amish, 2026-09-25 (PPR-DDR-002)

- **R3:** deflection judged at the 10 t maximum working force, with the molds closing on a metal stop (was 294 kN; status not met to met on paper).
- **R10:** target raised from \$720 to \$930. The decided bolted joints and hinged rail extension bring the BOM to \$984, so R10 was still not met; Amish topped up the budget to \$990 on 2026-09-26 (see below).
- **R11:** QC rack area relaxed from 0.8 x 0.5 m to 0.8 x 0.8 m; the press footprint is judged with the hinged rail extension folded (status not met to met on paper).
- **R9 and R12:** the pin-presence interlock and lead-free scrap are now part of the requirement text.

Requirement change decided by Amish, 2026-09-26

- **R10:** budget topped up from \$930 to \$990 ("I am ok with the budget top ups"; PPR-DDR-002 v0.2, item 16). The priced BOM of \$984 now meets R10 on paper with a \$6 (0.6 %) margin, which is thin; any price rise moves R10 back to at risk. The guarded version later the same day took the BOM to \$1,051 (see below).

Requirement change decided by Amish, 2026-09-26 (guarded version, PPR-DDR-003)

- **R9:** now states the guarding: fixed welded-mesh guards, a hinged front gate with a guard-locking interlock on the jack release, openings judged against ISO 13857, and controls outside the guard (status at risk to met on paper, with the ISO 13857 distances still an assumption).
- **R10:** the guards, gate and interlock add \$67, so the BOM is \$1,051 against \$990 (status met on paper to not met). The budget was left unchanged that day; Amish decided the overrun on 2026-09-27 (see below).
- **R11:** the guarded press is 940 x 700 mm, exactly at the 0.7 m depth limit (status unchanged, no margin).

Requirement change decided by Amish, 2026-09-27 (PPR-DDR-003 v0.2)

- **R10:** budget raised from \$990 to \$1,060 ("i agree with the budget for potpress"; option (a) in PPR-DDR-003). The priced BOM of \$1,051 meets R10 on paper with a \$9 (0.8 %) margin, which is thin; any price rise moves R10 back to at risk (status not met to met on paper).

Status after the design for construction, 2026-09-30 (PPR-DDR-004)

- **R10:** the parts a build needs (steel base plate under the female mold, thicker mold flanges, dowels and bushes, floor anchors, bolted jack plate, adapter disc, interlock details) bring the BOM to \$1,143 against \$1,060 (status within the value-engineering target to over it by \$83). The target is unchanged; the design decisions register lists the cost drivers and savings worth trying.
- No requirement text changed. R3, R8 and R11 figures are updated from PPR-CAL-001 v0.6; their status is unchanged.

■ Assumptions

- Pressing force needed for a well-consolidated wall is not published in the sources found (one low-cost press forms round-bottom filters with a 2 t jack, [Henry, Maley and Mehta, 2013](#)); 5 to 10 t is assumed as the working range and the 20 t jack rating is used as the design load (estimate, to be confirmed with a partner factory).
- Mix follows the RDI-C recipe reported by [Rayner \(2009\)](#); the reference filter needs about 7.5 kg of mix per pot, below the 8 to 9.5 kg factories report (PPR-CAL-001).
- The default flow acceptance band of 1.0 to 2.5 L/h follows Nicaraguan practice and the Potters for Peace range of 1.5 to 2.5 L/h; each factory sets its own band.
- Allowable stress in structural steel is taken as 165 MPa (S275 with a factor of about 1.67).
- Guard openings: 12.7 mm (1/2 in) square welded mesh, about 11 mm clear, with moving parts at least about 105 mm behind the mesh, is assumed to keep fingers out of the pinch zone. ISO 13857 is the reference; its tables were not checked here, so this is a stated assumption.